

Hierwave: the mathematics

Companion to `notes/reference.md` (October 2026)

The models, the sampler, and the training algorithm, as equations. Definitions only; the arguments are in `channels.tex` and the per-experiment notes. Section numbers follow `reference.md`.

1 Objects

Channels and views. A channel c at block size $h_c \in \{1, 2, 4, \dots\}$ holds $z_c : \Lambda_c \rightarrow \mathcal{D}_c = \{0, \dots, D_c - 1\}$ on the grid $\Lambda_c = \{0, \dots, H/h_c - 1\} \times \{0, \dots, W/h_c - 1\}$. A view is a map $\alpha : \mathcal{D}_c \rightarrow \{0, \dots, V_\alpha - 1\}$. A *fixed* mask $\phi_c \subseteq \Lambda_c$ marks cells the sampler never updates.

Parent cell. For $p \in \Lambda_a$ and a coarser-or-equal channel b ($h_a \leq h_b$),

$$q_b(p, \delta) = \left\lfloor \frac{p h_a}{h_b} \right\rfloor + \delta, \quad \delta \in \mathbb{Z}^2. \quad (1)$$

Factors. Each factor is a table E in energy units with $+\infty$ for hard rules.

$$\text{pair: } E_{a\alpha, b\beta, \delta}^{\text{pair}}(z) = \sum_{p \in \Lambda_a} T[\alpha(z_a(p)), \beta^{\text{pad}}(z_b(q_b(p, \delta)))], \quad (2)$$

$$\text{count: } E_{a\alpha, b\beta}^{\text{count}}(z) = \sum_{q \in \Lambda_b} T[\beta(z_b(q)), \sum_{p \in \text{block}(q)} \alpha(z_a(p))], \quad h_a < h_b, \quad (3)$$

$$\text{unary: } E_{a\alpha}^{\text{un}}(z) = \sum_{p \in \Lambda_a} T[\alpha(z_a(p))], \quad (4)$$

where β^{pad} returns the pad value when q is off the grid, and the term is dropped when the pad is -1 . A factor is *homed* on its finer side a ; the sampler of a channel reads only factors homed on it and reflections of same-level pairs (transposed table, negated offset). Top-down: no coarse channel reads a finer one during generation.

Certificate. On a tile channel with views $m = \text{mass}$, $\tau = \text{trunk}$, join bits $J_d(t, t') \in \{0, 1\}$ for the four sides d , and a distance channel $d_p \in \{0, \dots, D_{\text{max}}, \text{INF}\}$, cell p is valid iff

$$(m_p = 0 \wedge d_p = \text{INF}) \vee (\tau_p = 1 \wedge d_p = 0) \vee (d_p < \text{INF} \wedge \exists q \sim_d p : J_d(t_p, t_q) \wedge m_q \geq m_p \wedge d_q < d_p), \quad (5)$$

with energy $+\infty$ on an invalid cell and δd_p otherwise. With `tree`, exactly one neighbour satisfies the witness clause (the parent) and every other joined neighbour is a child ($m_q \leq m_p$, $d_q > d_p$).

Painted channels. A painter $\pi_c : \mathcal{D}_c^{\Lambda_c} \rightarrow \mathcal{D}_w^{\Lambda_w}$, $h_w < h_c$, writes a fine channel w deterministically from z_c (fixed everywhere). Honour factors read w by position-free pair tables; position dependence lives in π_c .

Promotion. For a view α of a large channel u , add a channel w with $\mathcal{D}_w = \text{range}(\alpha)$, move the factors that read $\alpha(z_u)$ to read w , and add

$$E^{\text{cell}}(w_p, u_p) = \text{cost}(w_p, \alpha(u_p)), \quad (6)$$

with cost either $\nu \mathbb{1}[w \neq \alpha(u)]$ (a tempering device), $-\log \hat{p}(w | u)$ from a corpus, or the free energy of Section 2. Marginalising w recovers the original model with its hard factors softened by exactly cost.

2 The two models

Levels $h_{\text{top}} > \dots > 1$; z_h the channels at level h ; $z_{>h}$ the coarser ones; $E_h(z_h | z_{>h})$ the sum of the factors homed on level h .

Bidirectional (the ideal).

$$p^*(z) = \frac{1}{Z} \exp \left(- \sum_h E_h(z_h | z_{>h}) \right). \quad (7)$$

Forward (the chain).

$$q(z) = \prod_h q_h(z_h | z_{>h}), \quad q_h = S_h \text{ Gibbs sweeps at } T = 1 \text{ under } E_h + \Phi_h \text{ from the painted initialisation}, \quad (8)$$

with $\Phi_h(z_h | z_{>h})$ the *learned potential* of level h .

The identity. Define recursively, from the bottom,

$$F_h(z_{>h}) = -\log \sum_{z_h} \exp \left(- E_h(z_h | z_{>h}) - F_{h/2}(z_h, z_{>h}) \right), \quad F_{1/2} \equiv 0. \quad (9)$$

Then $p^*(z_{\geq h}) \propto \exp \left(- \sum_{h' \geq h} E_{h'} - F_{h/2}(z_{\geq h}) \right)$, and the chain with equilibrated kernels reproduces p^* exactly iff

$$\Phi_h = F_{h/2} \quad \text{for every level } h. \quad (10)$$

The learned potential of a level is the free energy the levels below induce on it. With the budgeted kernel in place of exact sampling in the definition of F , the identity holds for the chain *as it runs* (the free energy as that kernel realises it at that budget).

Cluster expansion. $F_{h/2}$ is approximated by local terms over the cells of level h ,

$$F_{h/2}(z_h) \approx \sum_p f_1(z_p) + \sum_{\langle p, q \rangle} f_2(z_p, z_q) + \dots, \quad (11)$$

exact in the limit where the fine correlation length is below the block size; the residual of the fit (Section 4) measures the omitted orders.

Support and finite part. The support of Φ_h ($+\infty$ where no completion exists) comes from honourability checks, not from the fit; only the finite part is learned.

3 The sampler

At a non-fixed site p of the channel being swept, with candidate energies $e_p(t) = \sum_{\text{rows}} E(\cdot)$ evaluated with $z_p = t$ and everything else fixed,

$$t \sim \frac{\exp(-e_p(t)/T)}{\sum_{t'} \exp(-e_p(t')/T)} \quad \text{drawn as} \quad \arg \max_{t: e_p(t) < \infty} [-e_p(t)/T + G_t], \quad G_t \sim \text{Gumbel}(0, 1), \quad (12)$$

sites visited colour by colour $((x + y) \bmod 2$ at radius 1, $(x + 2y) \bmod 5$ at radius 2). With a certificate the draw is joint in (t, d) : for each t the valid d form an interval $[a_t, b_t]$ (above the smallest witness d , below the smallest d of a dependant), so

$$w_t = \exp(-e_p(t)/T) \sum_{d=a_t}^{b_t-1} e^{-\delta d/T} = \exp(-e_p(t)/T) e^{-\delta a_t/T} \frac{1 - e^{-\delta(b_t-a_t)/T}}{1 - e^{-\delta/T}}, \quad (13)$$

t by Gumbel-max over $\log w_t$, then d from the truncated geometric law on $[a_t, b_t]$.

4 The training (combining) algorithm

Fix a coarse channel c with reference value r , the corrected levels below it (their $E + \Phi$), and its painter π .

Windows. A window $W \subset \Lambda_c$ (e.g. 3×3) with halo ∂W (the one-cell ring). Draw z_W from a proposal (uniform, later cut from forward samples), $z_{\partial W}$ likewise, exterior at r . Let $z_W^r \equiv r$ on W with the same halo. The fine region $R(W)$ is the blocks of $W \cup \partial W$ plus the painters' spill.

Free energy per window.

$$F(z_W \mid z_{\partial W}) = -\log \sum_{z_{R(W)}} \exp(-E_{\text{fine}}(z_{R(W)} \mid z_W, z_{\partial W})), \quad y(W) = F(z_W \mid z_{\partial W}) - F(z_W^r \mid z_{\partial W}). \quad (14)$$

Estimators: exact hooks (transfer matrix; per-cell products when the fine cells are independent given the paint), or annealed importance sampling with the fine kernel. Split $E_{\text{fine}} = E_{\text{un}} + E_{\text{rest}}$ (unaries of the fine channel vs. everything else, with $+\infty \rightarrow L$ in E_{rest}), $\beta_0 = 10^{-3} < \beta_1 < \dots < \beta_K = 1$,

$$\log Z_0 = \sum_{s \in R(W)} \log \sum_t e^{-E_{\text{un},s}(t)}, \quad (15)$$

$$x_0 \sim p_0 \propto e^{-E_{\text{un}}}, \quad x_k \sim \text{one Gibbs sweep of } p_{\beta_k} \propto e^{-E_{\text{un}} - \beta_k E_{\text{rest}}} \text{ from } x_{k-1}, \quad (16)$$

$$\log w = -\sum_{k=0}^{K-1} (\beta_{k+1} - \beta_k) E_{\text{rest}}(x_k), \quad \widehat{\log Z} = \log Z_0 + \log \text{meanexp}_{m=1..M} \log w^{(m)}, \quad \hat{F} = -\widehat{\log Z}. \quad (17)$$

$\hat{F}$ is biased upward at small K ; $K = 256$, $M = 4-8$ was adequate on the toys.

Paint potential. Over the painted channel(s) at fine resolution with values $\{0, \dots, V-1\}$ and the four independent offsets $\mathcal{O}_8 = \{(0, 1), (1, 0), (1, 1), (1, -1)\}$,

$$F_\theta(\pi) = \sum_t u(\pi_t) + \sum_{\delta \in \mathcal{O}_8} \sum_t g_\delta(\pi_t, \pi_{t+\delta}), \quad \theta = (u, \{g_\delta\}), \quad (18)$$

off-grid partners dropped. F_θ is linear in θ through the count features $\Phi(\pi) \in \mathbb{R}^{V+|\mathcal{O}_8|V^2}$. Regression with the halo painted:

$$x(W) = \Phi(\pi(z_W, z_{\partial W})) - \Phi(\pi(z_W^r, z_{\partial W})), \quad \hat{\theta} = \arg \min_{\theta} \sum_W (x(W) \cdot \theta - y(W))^2 + \lambda \|\theta\|^2. \quad (19)$$

The halo's own terms cancel in x and y ; the window-halo cross terms are features. The relative residual $\sqrt{\sum (x \cdot \hat{\theta} - y)^2 / \sum y^2}$ measures what is not local in the painted map at this order (and the AIS noise floor).

Materialisation. For the forward kernel's tables over $\mathcal{D}_c$, with $\pi(\cdot)$ on a background of r ,

$$u_c(v) = F_{\hat{\theta}}(\pi(v)) - F_{\hat{\theta}}(\pi(r)), \quad (20)$$

$$g_{c,d}(v, v') = F_{\hat{\theta}}(\pi(v \text{ at } p, v' \text{ at } p+d)) - u_c(v) - u_c(v') - F_{\hat{\theta}}(\pi(r)), \quad (21)$$

for the coarse adjacencies d (east, south, diagonals). Exact for a linear F_θ . For large D , evaluate $F_{\hat{\theta}}$ on the candidate's painted footprint at sampling time instead. Upward recursion: the next coarser channel uses c with $E_c + \Phi_c$ as its fine level, $p_0 =$ the unaries of c (designed and learned).

Gauge. Only y is observed, so u and $\{g_\delta\}$ are identified up to the transformations that leave every window sum invariant; materialised coarse tables are invariant and are what is used. When comparing coarse tables, fix the reference gauge ($g_c(r, \cdot) = g_c(\cdot, r) = 0$) or double-centre.

5 The conv potential (coarse-level pair potentials)

For a channel with value embedding $e : \mathcal{D} \rightarrow \mathbb{R}^k$ (rows of $E \in \mathbb{R}^{V \times k}$), offsets $d \in \{-1, 0, 1\}^2$,

$$E(z) = \sum_p \left[a \cdot e(z_p) + \sum_{d \neq 0} e(z_p)^\top A_d e(z_{p+d}) + v \cdot \text{softplus} \left(\sum_d W_d e(z_{p+d}) + b \right) \right], \quad (22)$$

$A_d \in \mathbb{R}^{k \times k}$, $W_d \in \mathbb{R}^{k \times m}$, $b, v \in \mathbb{R}^m$; off-grid neighbours use $e(\text{pad})$ or drop out. With one-hot E and $m = 0$ this is the 8-offset pair-table model and the fit is closed-form ridge; $m > 0$ (the plaquette head) adds 3- and 4-body terms, fitted by Adam. Candidate cost in the kernel: $9k$ (bilinear) + $9km$ (head, incremental). On the toys the head added nothing measurable; the bilinear form reached the estimator's noise floor.

6 Self-play moment matching (retained as a diagnostic)

For learned tables with sufficient statistics $N_k(z)$ (counts of table entries), a forward sample $z \sim q$, and a one-site kernel that leaves p^* invariant,

$$\Delta_k(z) = \sum_p \sum_t p^*(z_p = t \mid z_{-p}) [N_k(z \text{ with } z_p = t) - N_k(z)], \quad \theta_k \leftarrow \theta_k - \eta \mathbb{E}_q[\Delta_k], \quad (23)$$

the Rao-Blackwellised one-step correction; CD- K replaces the bracket by $N_k(z_K) - N_k(z) + \Delta_k(z_K)$ after K sweeps of the p^* kernel. Fixed point: q invariant under the kernel on the features. Under hard honour the coarse conditional is a point mass and $\Delta \equiv 0$ (frozen sites), which is why the free-energy route replaced it.

7 The toys' exact quantities

Circles (tiles independent given the paint). Per tile with demand π_t , field μ on air, honour κ per violation:

$$Z_{\text{free}} = 1 + e^{-\mu}, \quad Z_{\text{dirt}} = 1 + e^{-\mu-\kappa}, \quad Z_{\text{air}} = e^{-\mu} + e^{-\kappa}, \quad Z_{\text{conflict}} = e^{-\kappa} + e^{-\mu-\kappa}, \quad (24)$$

$F(\text{objects}) = -\sum_t \log Z_{\pi_t}$, hence a conflict tile costs $-\log Z_{\text{conflict}} + \log Z_{\text{dirt}} + \log Z_{\text{air}} - \log Z_{\text{free}}$ and a shared ring tile gains $\log Z_{\text{air}} - \log Z_{\text{free}}$. The induced mid potential is exactly a unary paint potential ($g_\delta \equiv 0$); the presence bonus $b = F_{\text{obj}} - \log(\#\text{offsets})$ makes an isolated block 50/50.

Potts (q colours, cyclic distance d , tile coupling J , honour κ , palettes with λ). At $\kappa \rightarrow \infty$ the induced block coupling is $g(m, m') = B J d(m, m')$ per shared edge of length B ; the mid level is a q -state clock model with coupling BJ , ordered for $J \gtrsim 0.2$ at $B = 4$, $q = 4$. Block free energies by a column transfer matrix over q^B states.